

Vector Database Selection and Comparative Analysis

Project: MediAssist-AI

Healthcare Document Intelligence System using Retrieval Augmented Generation (RAG)

1. Purpose

The objective of this document is to evaluate and compare multiple Vector Database (VectorDB) solutions for the MediAssist-AI project and justify the final technology selection.

The comparison focuses on the following evaluation parameters:

1. Indexing Algorithm
2. Search Capabilities
 - Keyword Search
 - Vector Search
 - Hybrid Search
3. Data Scale Support (< 1 Million Vectors)

The Vector Databases evaluated are:

- FAISS
 - ChromaDB
 - Pinecone
 - Weaviate
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2. Project Requirements

The MediAssist-AI system is designed to process:

- Medical Reports
- Discharge Summaries
- Prescriptions
- Lab Reports
- Healthcare Guidelines

The expected vector volume is below one million vectors.

Estimated Scale

Parameter	Expected Value
Documents	5,000 – 20,000
Chunks per Document	20 – 100
Total Vectors	100,000 – 500,000
Scale Category	Less than 1 Million Vectors

Since the system is initially intended for local development and academic evaluation, the selected VectorDB should be lightweight, easy to integrate, and cost-effective.

3. Comparative Analysis

3.1 FAISS

Overview

FAISS (Facebook AI Similarity Search) is an open-source vector similarity search library developed by Meta AI.

Indexing Algorithms

FAISS supports multiple indexing strategies:

- Flat Index (Exact Search)
- IVF (Inverted File Index)
- HNSW
- Product Quantization (PQ)

Commonly used index:

IndexFlatL2

Search Capabilities

Search Type	Support
Keyword Search	No
Vector Search	Yes
Hybrid Search	Custom Implementation Required

Data Scale Support

Scale	Support
< 1 Million Vectors	Excellent
1M – 10M Vectors	Excellent
>10M Vectors	Supported with optimized indexing

Advantages

- Open Source
- High Performance
- Fast Similarity Search
- Local Deployment
- Strong LangChain Integration
- No Infrastructure Cost

Limitations

- No Native Metadata Filtering
- No Built-in Keyword Search
- Hybrid Search Requires Additional Components

3.2 ChromaDB

Overview

ChromaDB is an open-source vector database designed specifically for LLM and RAG applications.

Indexing Algorithm

Internally uses:

HNSW (Hierarchical Navigable Small World)

Search Capabilities

Search Type	Support
Keyword Search	Limited
Vector Search	Yes
Hybrid Search	Partial Support

Data Scale Support

Scale	Support
< 1 Million Vectors	Very Good
1M – 2M Vectors	Good
>2M Vectors	Performance May Degrade

Advantages

- Easy Setup
- Built-in Persistence
- LangChain Native Integration
- Simple Development Experience

Limitations

- Limited Search Flexibility
- Less Configurable than FAISS
- Not Optimized for Very Large Datasets

3.3 Pinecone

Overview

Pinecone is a fully managed cloud-native vector database platform designed for production-scale AI systems.

Indexing Algorithm

Uses proprietary Approximate Nearest Neighbor (ANN) indexing optimized for distributed search.

Search Capabilities

Search Type	Support
Keyword Search	Yes
Vector Search	Yes
Hybrid Search	Yes

Data Scale Support

Scale	Support
< 1 Million Vectors	Excellent
Millions of Vectors	Excellent
Billions of Vectors	Excellent

Advantages

- Enterprise Grade
- Fully Managed Infrastructure
- Native Hybrid Search
- High Availability
- Automatic Scaling

Limitations

- Paid Service
- Cloud Dependency
- Vendor Lock-In
- Not Ideal for Academic Projects

3.4 Weaviate

Overview

Weaviate is an open-source vector database that combines vector search with keyword search and metadata filtering.

Indexing Algorithm

Uses:

HNSW (Hierarchical Navigable Small World)

Search Capabilities

Search Type Support

Keyword Search Yes

Vector Search Yes

Hybrid Search Yes

Data Scale Support

Scale Support

< 1 Million Vectors Excellent

Millions of Vectors Excellent

Enterprise Scale Excellent

Advantages

- Native Hybrid Search
- GraphQL Support
- Metadata Filtering
- Enterprise Features
- Open Source

Limitations

- Complex Deployment
- Higher Infrastructure Requirements
- Steeper Learning Curve

4. Comparison Matrix

Parameter	FAISS	ChromaDB Pinecone		Weaviate
Indexing Algorithm	Flat, IVF, HNSW, PQ	HNSW	Proprietary ANN	HNSW
Keyword Search	No	Limited	Yes	Yes
Vector Search	Yes	Yes	Yes	Yes
Hybrid Search	Custom	Partial	Yes	Yes

Parameter	FAISS	ChromaDB Pinecone		Weaviate
Metadata Filtering	Limited	Yes	Yes	Yes
Local Deployment	Yes	Yes	No	Yes
Cloud Deployment	Manual	Manual	Native	Native
Open Source	Yes	Yes	No	Yes
LangChain Integration	Excellent	Excellent	Excellent	Excellent
Performance (<1M Vectors)	Excellent	Very Good	Excellent	Excellent
Infrastructure Complexity	Low	Low	Very Low	Medium
Cost	Free	Free	Paid	Free / Enterprise

5. Analysis Based on Project Requirements

The MediAssist-AI project has the following characteristics:

- RAG-based document retrieval system
- Less than one million vectors
- Academic and capstone implementation
- Local development environment
- Fast prototyping requirement
- Minimal infrastructure overhead

Evaluation Summary

FAISS

Best suited for local RAG implementations with moderate data volume. Provides excellent retrieval performance and low infrastructure complexity.

ChromaDB

Good alternative for quick development but offers fewer indexing choices and flexibility compared to FAISS.

Pinecone

Technically powerful but introduces unnecessary cloud dependency and operational costs for the current project scope.

Weaviate

Feature-rich and enterprise-ready, but deployment complexity exceeds the requirements of the project.

6. Final Recommendation

After evaluating all candidate Vector Databases against the defined criteria, FAISS is selected for MediAssist-AI.

Selected Vector Database

FAISS

Justification

1. Excellent performance for datasets below one million vectors.
 2. Multiple indexing algorithms for future scalability.
 3. Open-source and free to use.
 4. Easy integration with LangChain and RAG pipelines.
 5. Minimal infrastructure requirements.
 6. Suitable for local development and academic projects.
 7. Fast semantic similarity search.
 8. No cloud dependency or recurring operational cost.
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7. Conclusion

FAISS provides the most appropriate balance of performance, simplicity, scalability, and cost-effectiveness for the MediAssist-AI healthcare document intelligence platform. Considering the project's expected vector volume, local deployment requirements, and academic objectives, FAISS is selected as the primary vector database for implementation.